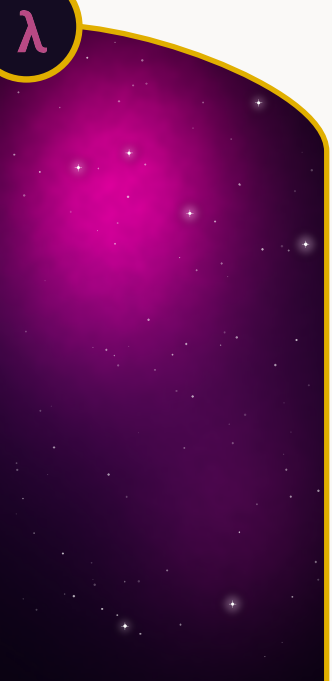


L9A: Environment Model

CS1101S: Programming Methodology

Martin Henz

14th October, 2026

Version 1.1 — Built: 2026-10-13 23:29:10 SGT

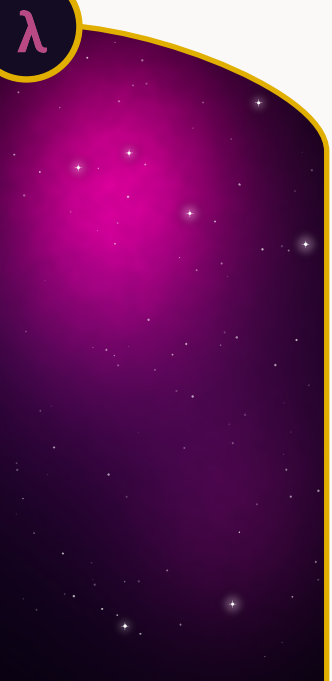
Reading Assessment 2

- Scope: All material up to and including Week 9's Lectures, Paths, Reflections, Studios and Studio S10.
- Format: similar to RA1; look out for our announcements
- Time: Friday, 23/10 (Week 10), 10:00am
- Venues: look out for our announcements

Sumobot 2026: Only the best will win!

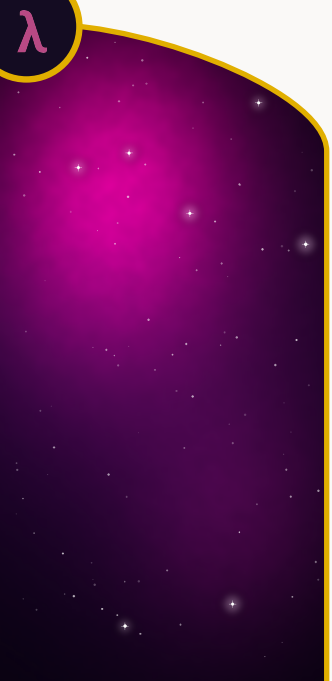
- October 15th, 6pm
- Dinner, prizes, swag, and a guest talk about Jane Street.
- Robot talent contest as well.





State (3.1)

Environment Model (3.2 and 3.3)



State (3.1)

Environment Model (3.2 and 3.3)

Recap: Assignment

- The **declaration assignment** gives `x` its first value.

```
x = 1
```



Recap: Assignment

- The **declaration assignment** gives `x` its first value.
- The **reassignment** evaluates the right-hand side, then changes the binding of `x`.

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x = x + 1
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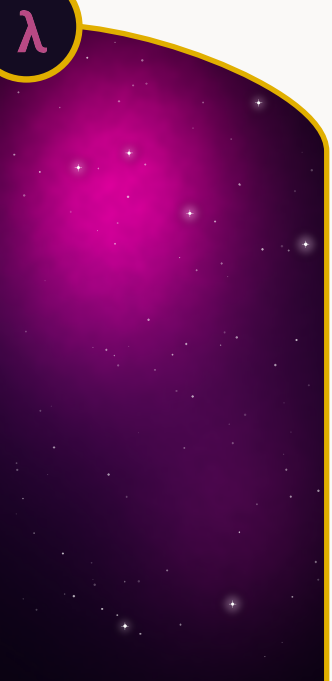


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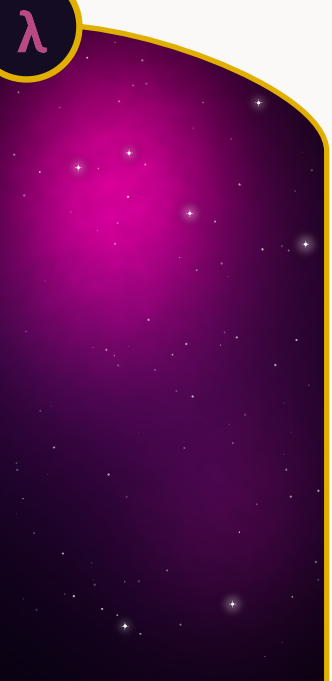
- The **declaration assignment** gives `x` its first value.
- The **reassignment** evaluates the right-hand side, then changes the binding of `x`.
- `x` now evaluates to 2.

```
x = 1
x = x + 1
print(x)
```



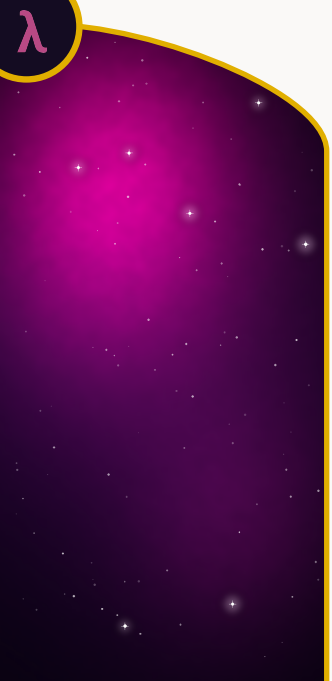


Recap: Reassignment: Pros



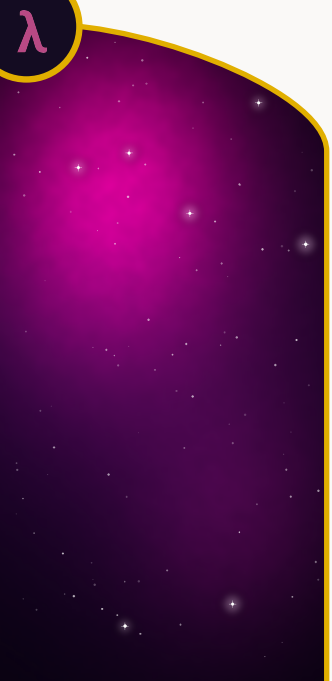
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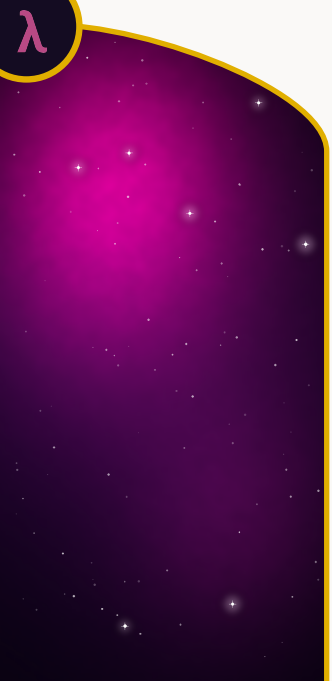


Recap: Reassignment: Pros

- Reassignment allows us to create objects with state.
- State allows objects to behave differently over time.

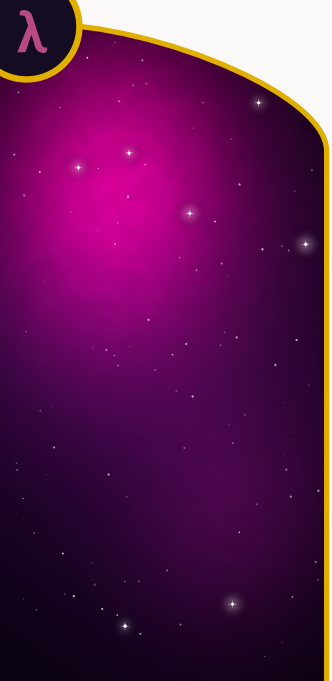


Recap: Reassignment: Cons



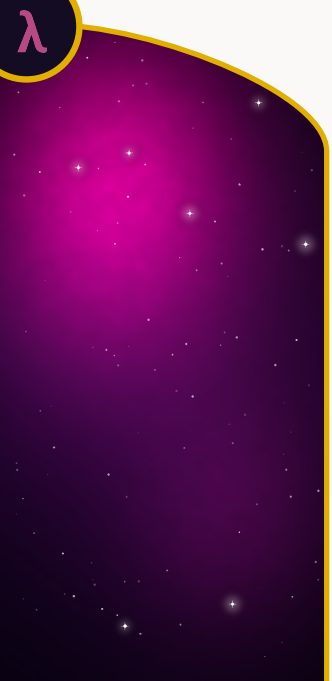
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  - Harder to verify correctness



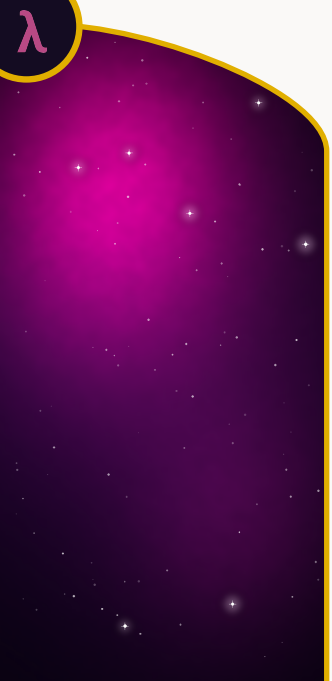
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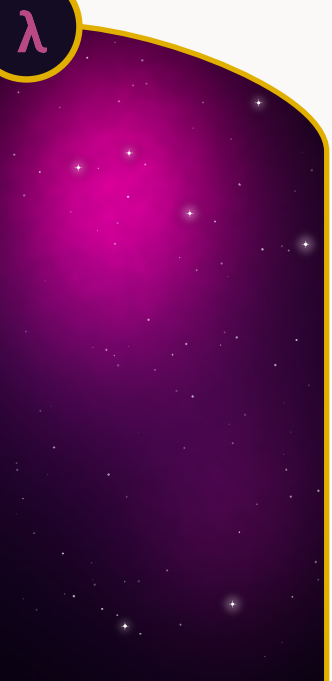
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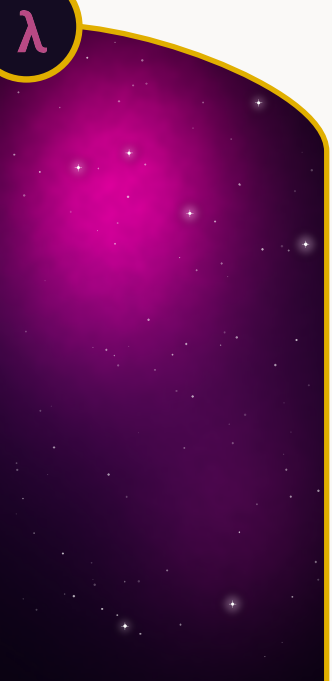


Recap: Reassignment: Cons

- Harder to reason about programs
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- The substitution model of evaluation breaks down!
  - It is not powerful enough to explain state.
  - We need a more sophisticated model: the environment model.

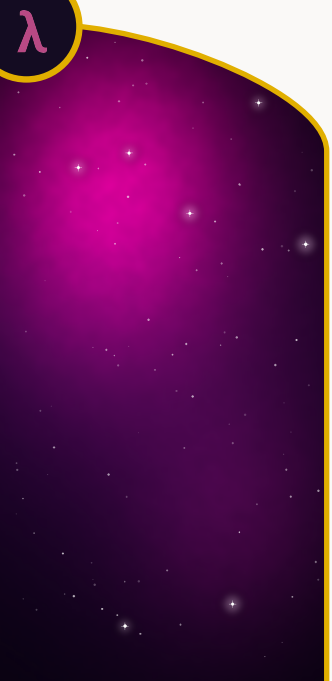


Recap: Why Substitution Model Breaks Down?



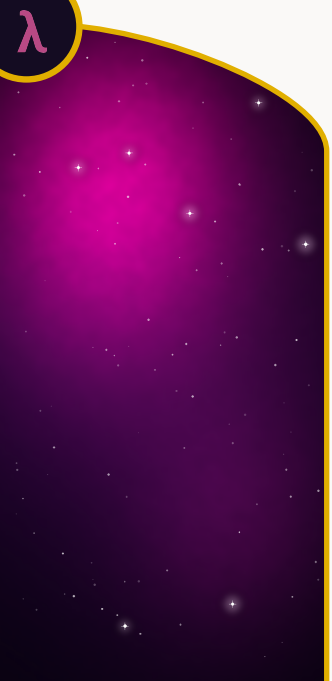
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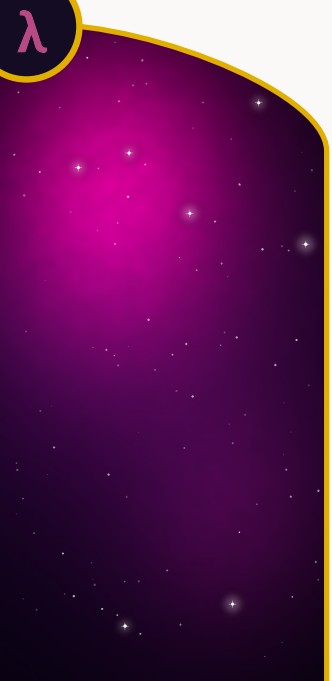
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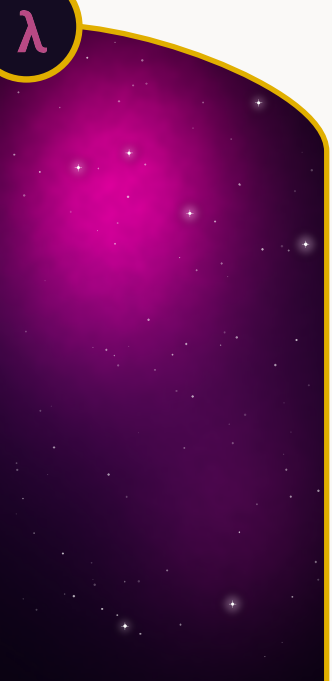
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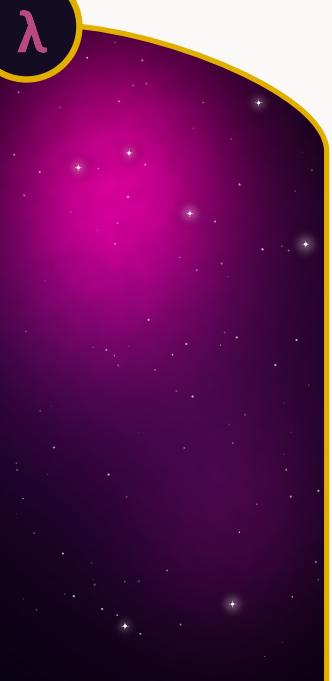
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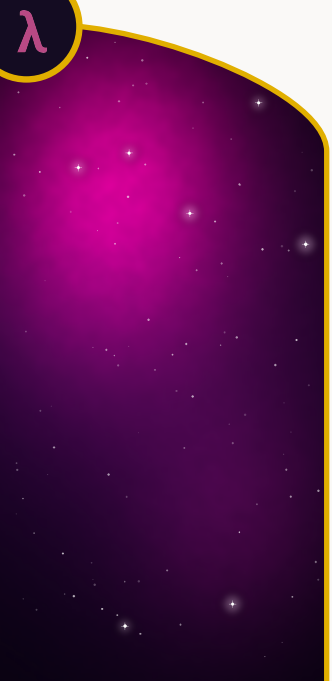
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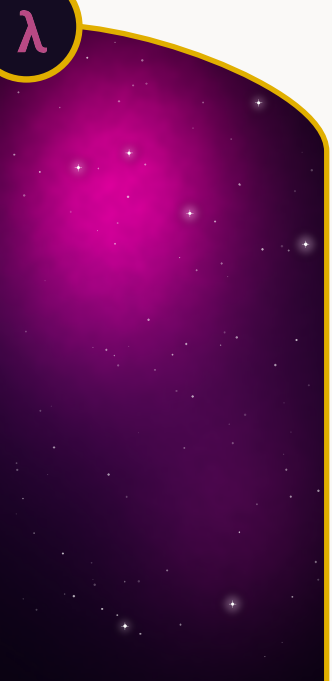


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  - The contents of the container may change over time.
  - The container is maintained in a structure called an environment.



Environments & Environment Model



Environments & Environment Model

- Environments

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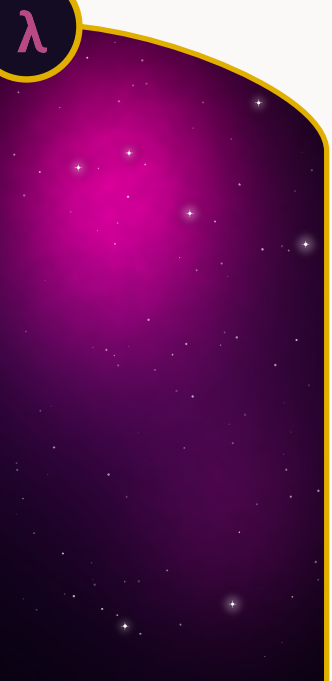
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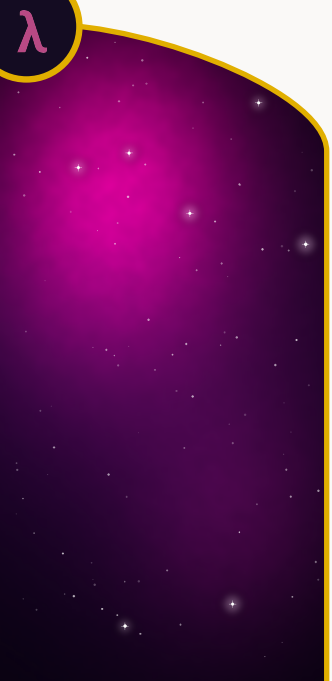
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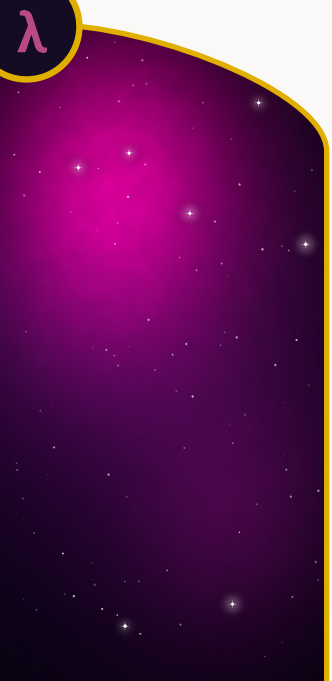
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  - A mental model for environments
  - A set of rules for managing environments
  - A graphical representation: environment diagrams

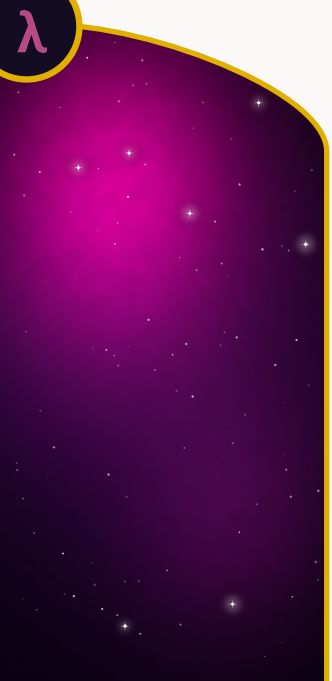


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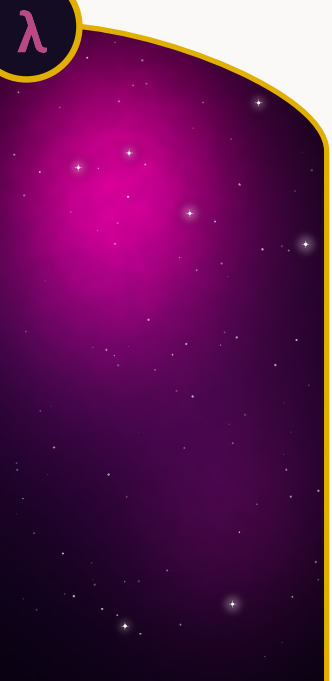


Environments: Concept



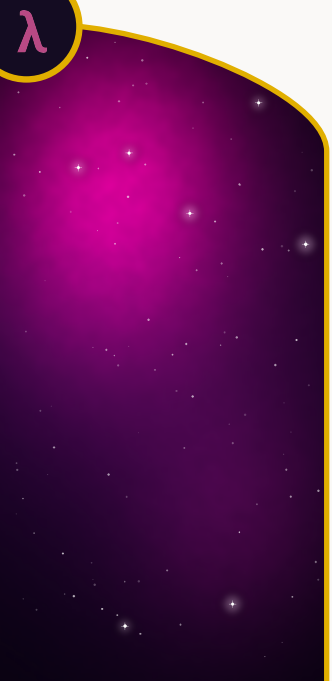
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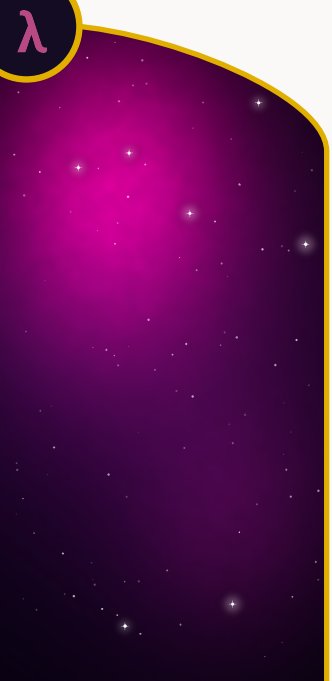


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- Every expression is evaluated with respect to the **current** environment.

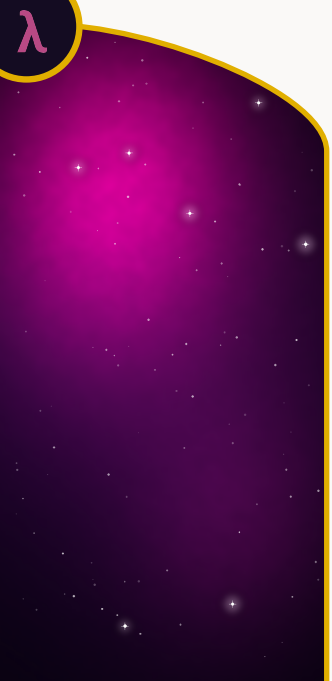


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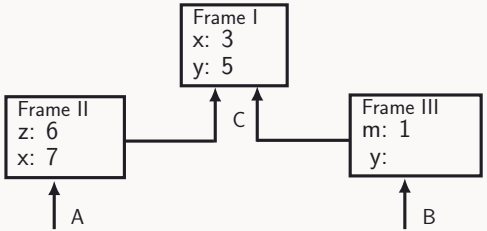
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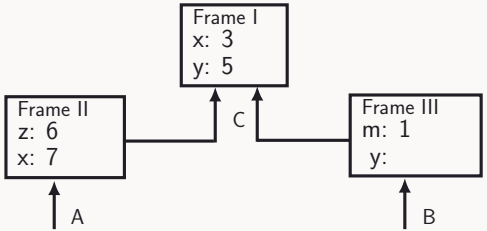
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- Extending an environment means adding a new frame "inside" it.

Example Environments

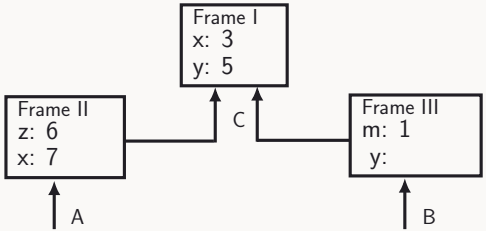


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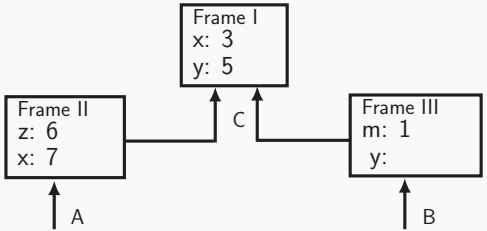
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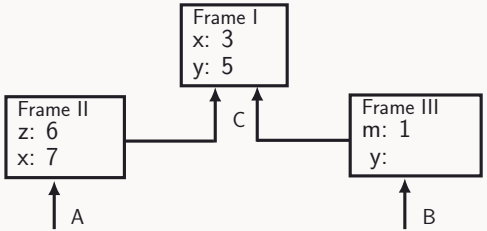
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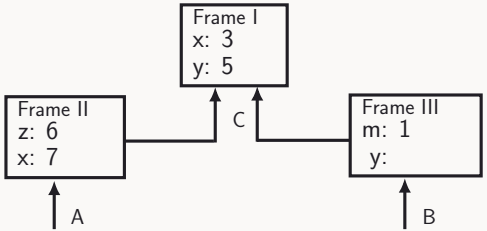
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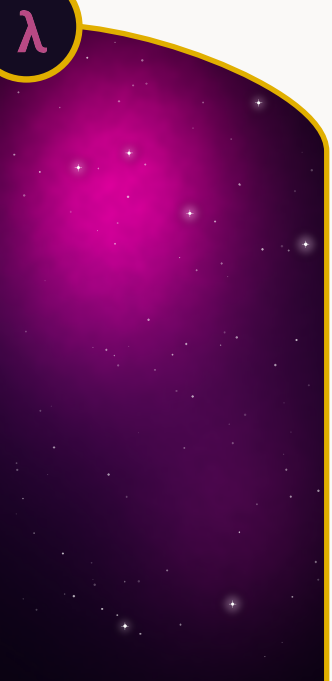


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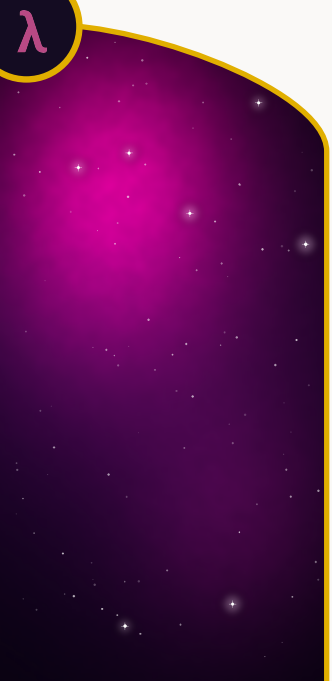
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Accessing an Environment

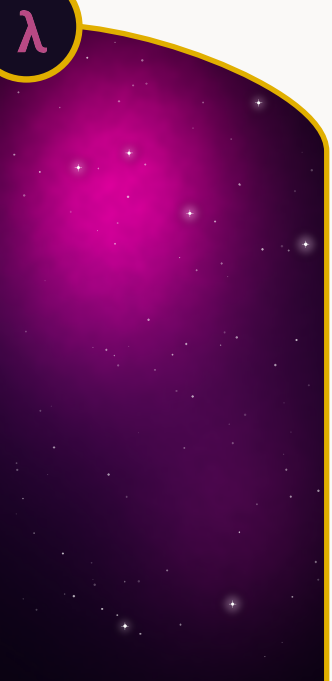


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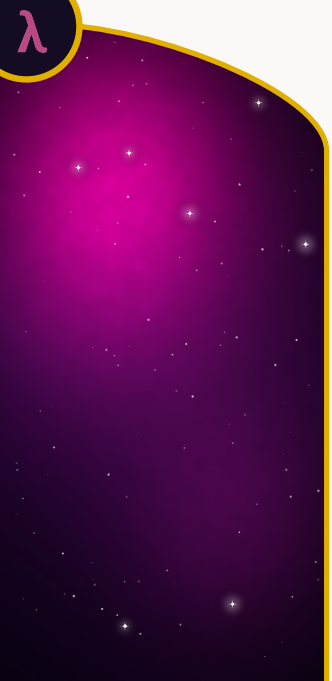
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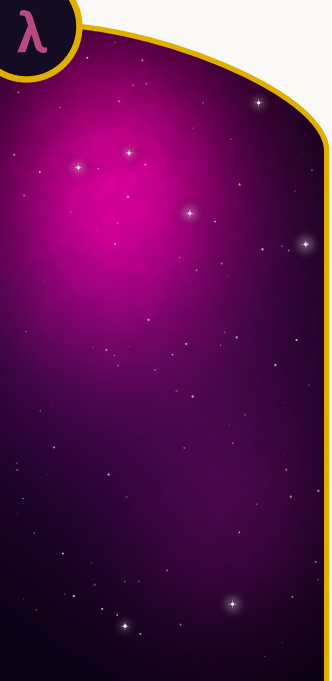


Built-ins Environment



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- The built-ins environment is one frame containing built-ins and pre-declared names.

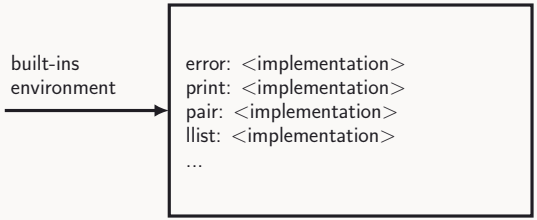


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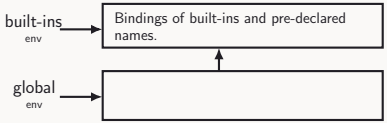
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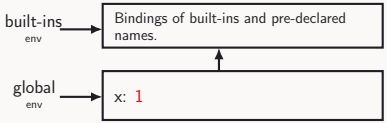
Example: Top-Level Assignments

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print(x)
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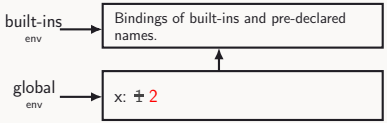
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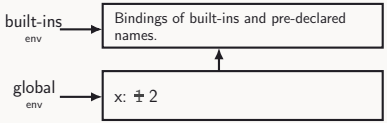
```
x = 1
x = x + 1
print(x)
```

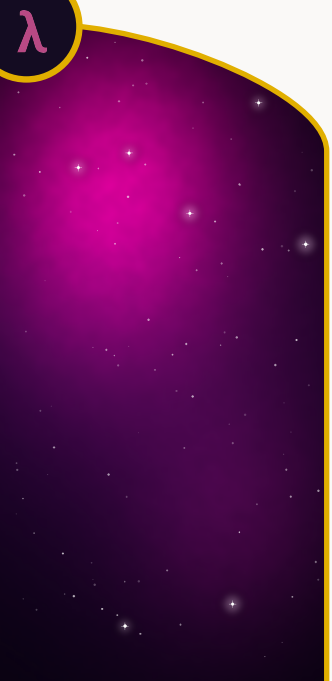


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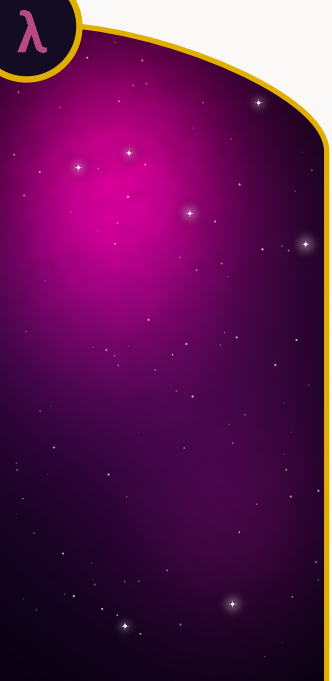
output: 2





The Global Environment

- Evaluating the user program creates the global environment.



The Global Environment

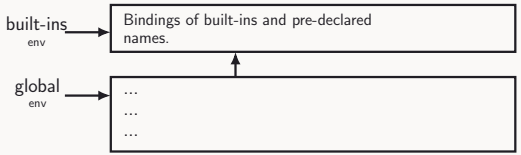
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  - Global declaration assignments add bindings to the global frame.

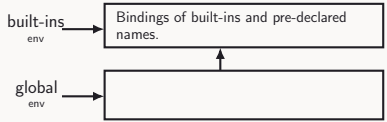


Example: A Function Call Creates a Frame

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x = 3

def fun():
    x = 5
    y = 8

fun()
```

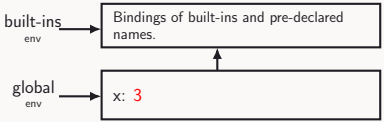


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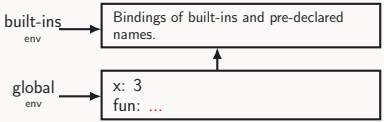


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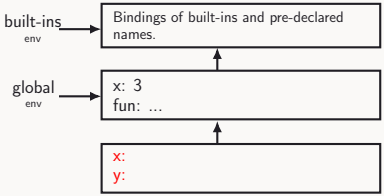


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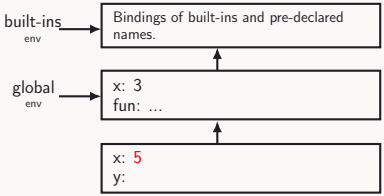


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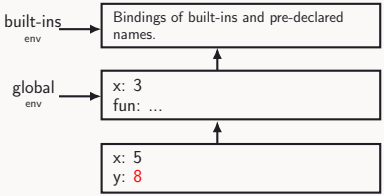


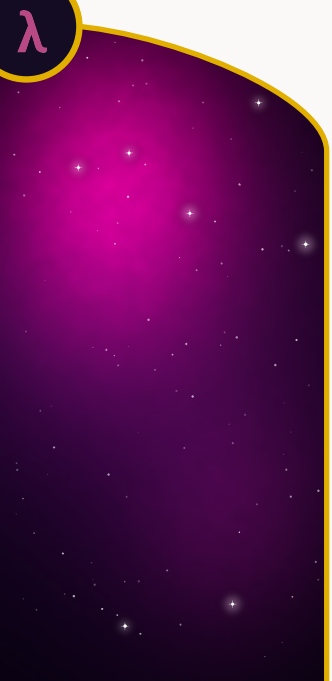
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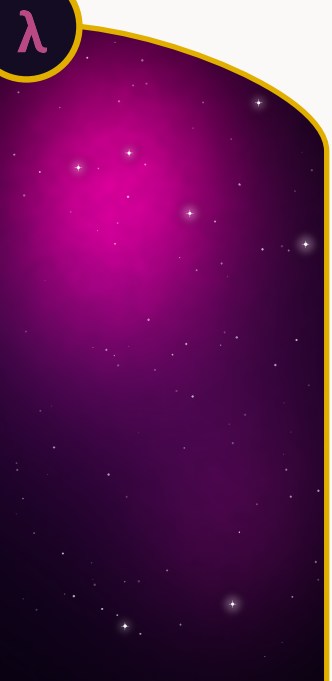
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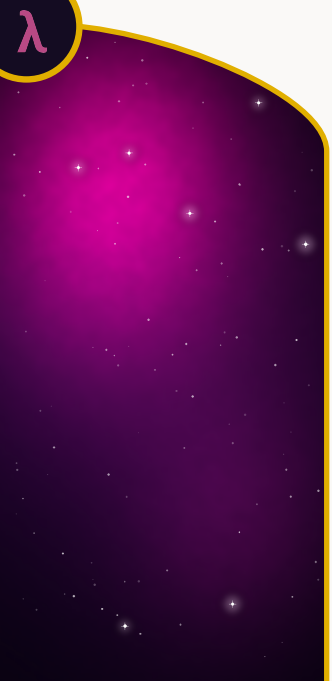


Frame of a Function Call



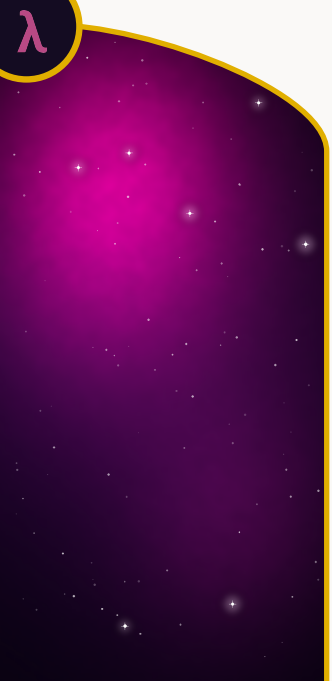
Frame of a Function Call

- Start with the environment stored in the function object.



Frame of a Function Call

- Start with the environment stored in the function object.
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  - Bind each local variable, initially unassigned.



Frame of a Function Call

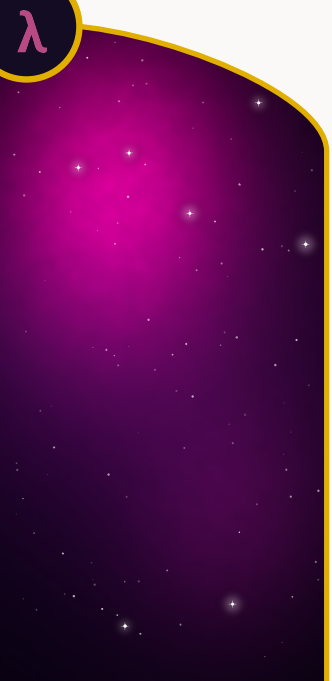
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- Evaluate the function body in that environment.

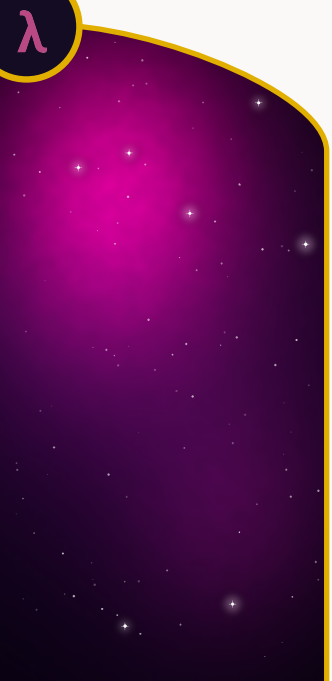
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- Local variables are the names that occur in any assignment in the body, except if the name is declared global or nonlocal in that body.
- Evaluate the function body in that environment.
- For clarity, empty frames may be omitted from our diagrams.



Evaluation of Assignments

- When a function body is entered, its local names are placed in the call frame, initially unassigned.

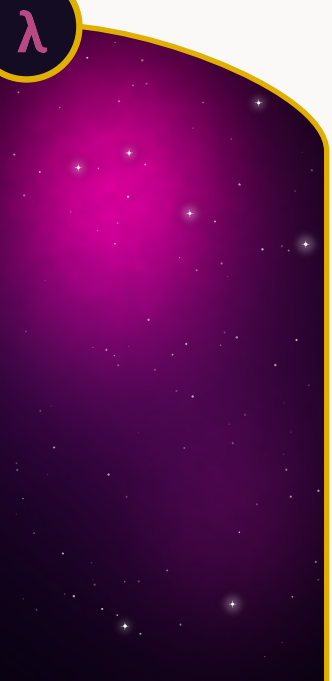


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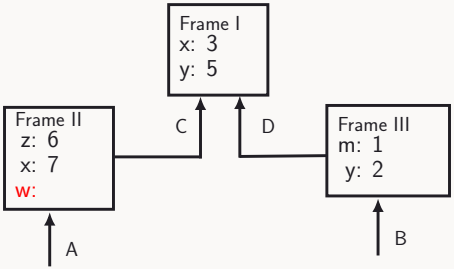
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- When a function body is entered, its local names are placed in the call frame, initially unassigned.
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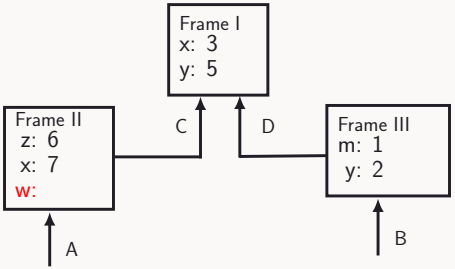
Evaluation of Assignments

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- To evaluate `name = expression`:
  - Evaluate `expression`.
  - Assign the resulting value to the selected binding.
- Reading a name before it has been assigned signals an error.

Example Environments

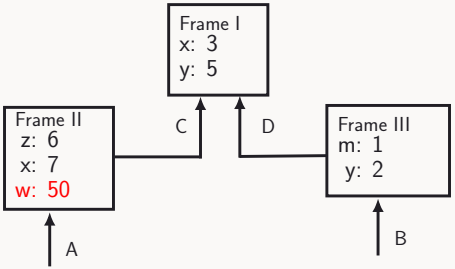


Example Environments



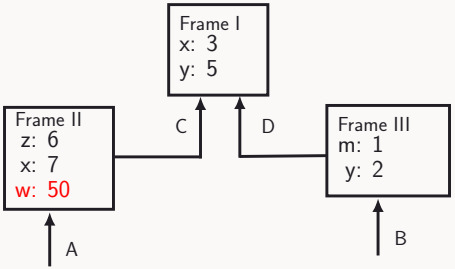
- Evaluate in A: `w = 50`.

Example Environments



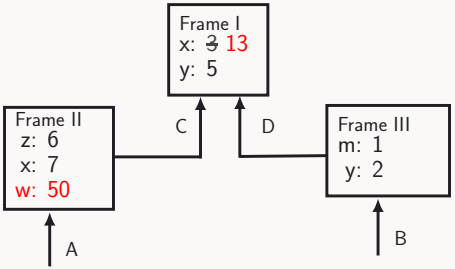
- Evaluate in A: `w = 50`.
- `w` in Frame II is assigned the value 50.

Example Environments



- Evaluate in B: `x = x + 10`.

Example Environments



- Evaluate in B: `x = x + 10`.
- The value of `x` in Frame I changes from 3 to 13.

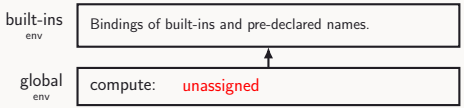
Function-Call Trace

- The trace focuses on these nested function calls:

```
def outer():
    x = 5
    y = 8
    def inner():
        x = 20 + y
        return x
    inner()
    x = x + 1
    return x
return outer()
```



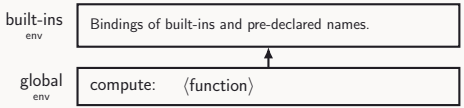
Environment Model Trace




```
def compute(x):
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compute(3)
```

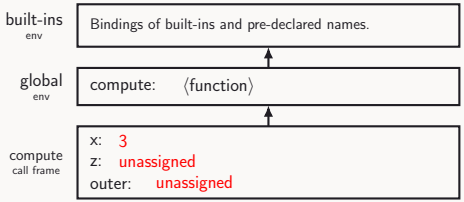
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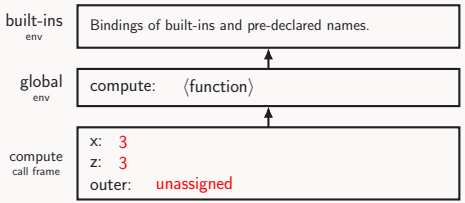
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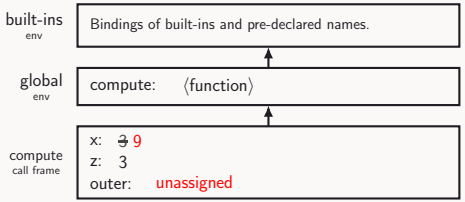
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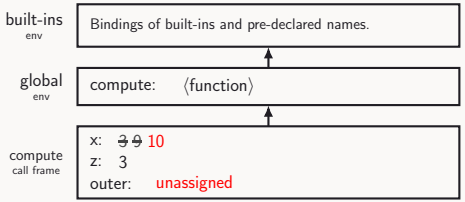
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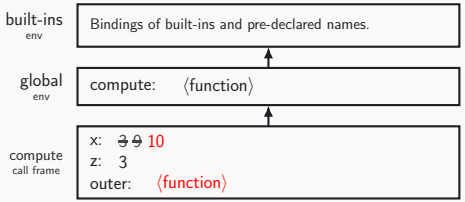
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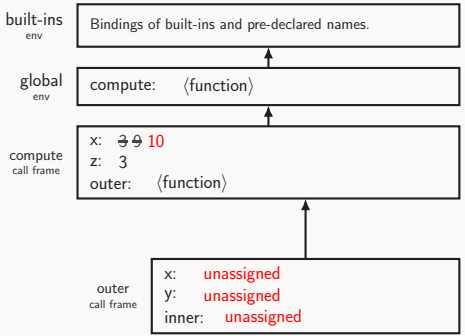
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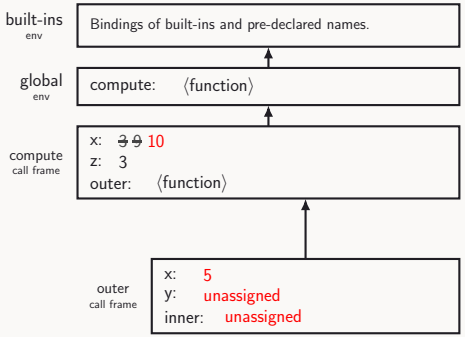
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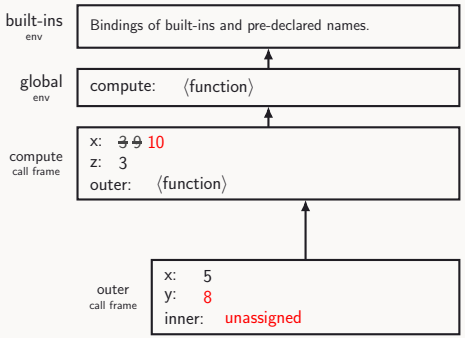
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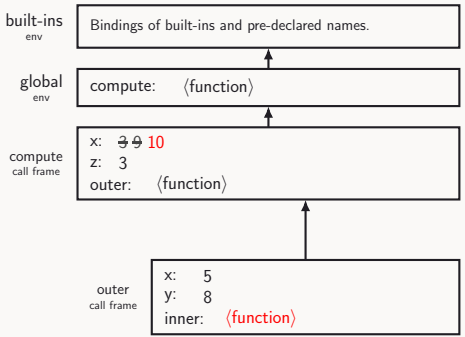
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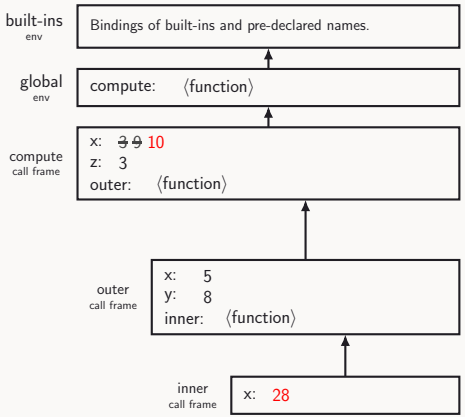
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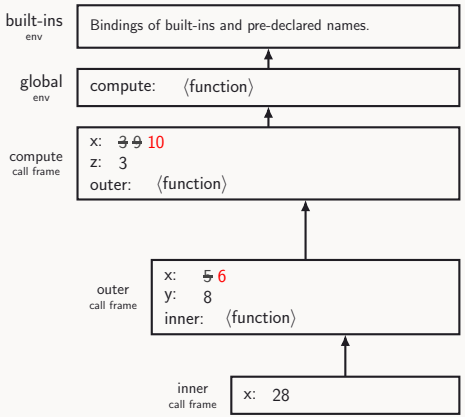
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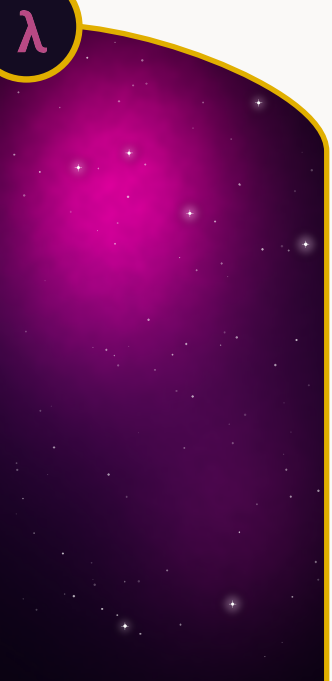
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Evaluation of Lambda Expressions

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  - One circle points to the function's parameters and body.
  - The other points to the environment in which the lambda was evaluated.
  - The function therefore remembers the environment in which it was created.

Example for Evaluating Lambda Expressions

```
square = lambda x: x * x
print(square(5))
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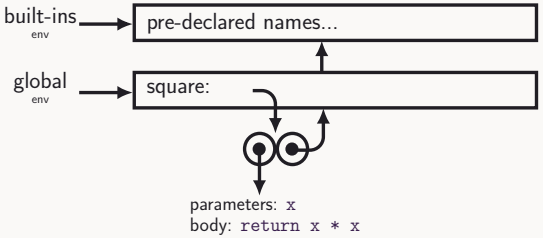


- After evaluating the assignment in the program environment:

Example for Evaluating Lambda Expressions

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```

- After evaluating the assignment in the program environment:



Example for Evaluating Function Definitions

```
def square(x):
    return x * x
```



Example for Evaluating Function Definitions

```
def square(x):
    return x * x
```

which is equivalent to

```
square = lambda x: x * x
```

Example for Evaluating Function Definitions

```
def square(x):
    return x * x
```

which is equivalent to

```
square = lambda x: x * x
```

- After evaluating the function definition in the program environment:

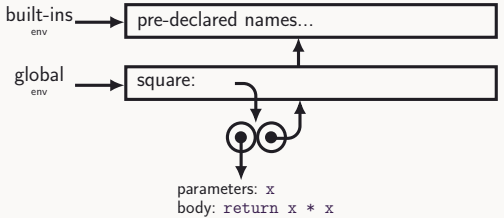
Example for Evaluating Function Definitions

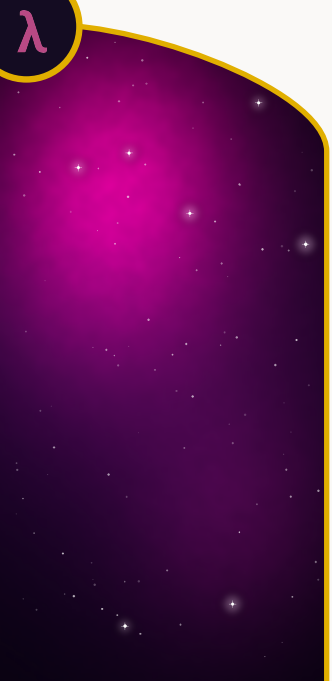
```
def square(x):
    return x * x
```

which is equivalent to

```
square = lambda x: x * x
```

- After evaluating the function definition in the program environment:





Evaluation of Function Applications

Evaluation of Function Applications

- To evaluate a function application:

  ```
  fun(arg1, arg2, ..., argn)
  ```

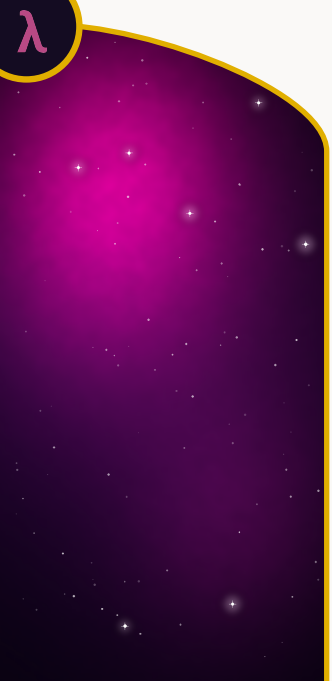
  - Evaluate the subexpressions in the current environment.

Evaluation of Function Applications

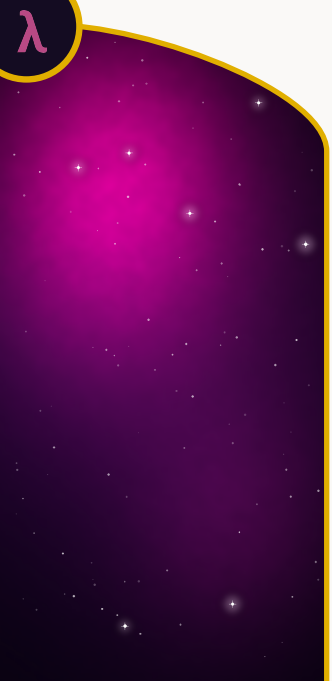
- To evaluate a function application:

```
fun(arg1, arg2, ..., argn)
```

  - Evaluate the subexpressions in the current environment.
  - Apply the value of `fun` to the values of the argument subexpressions.

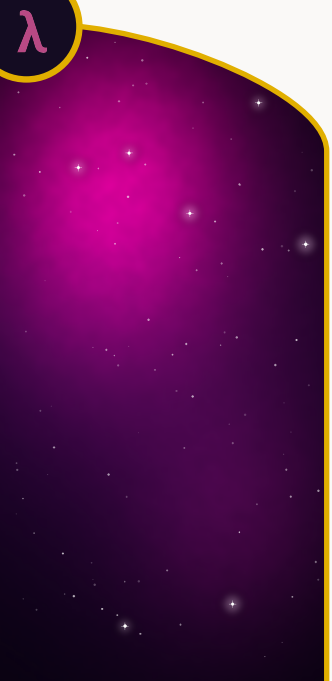


Applying a Function Value to Argument Values



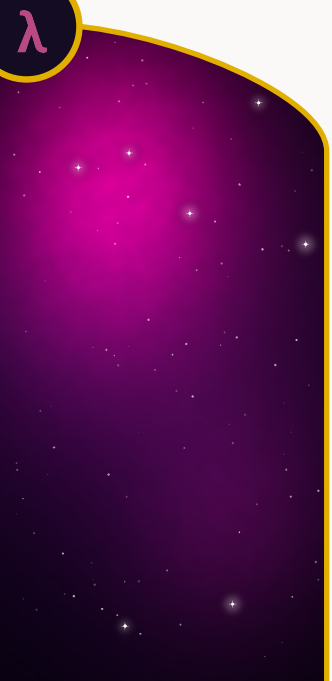
Applying a Function Value to Argument Values

- Take the environment stored in the function object.



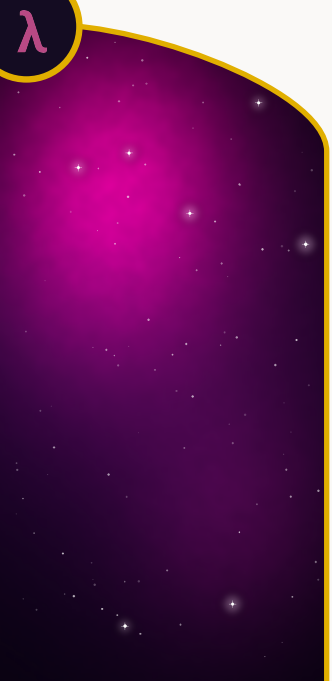
Applying a Function Value to Argument Values

- Take the environment stored in the function object.
- Extend it with a call frame.



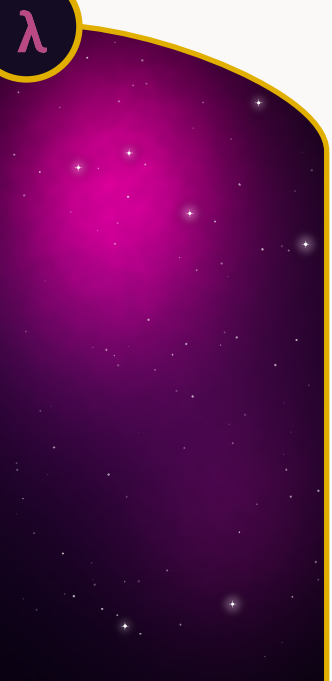
Applying a Function Value to Argument Values

- Take the environment stored in the function object.
- Extend it with a call frame.
- Bind the parameters to the argument values.



Applying a Function Value to Argument Values

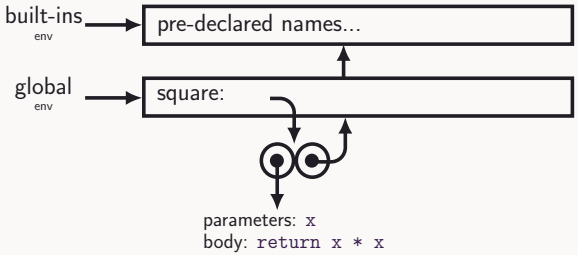
- Take the environment stored in the function object.
- Extend it with a call frame.
- Bind the parameters to the argument values.
- Bind each local variable, initially unassigned.



Applying a Function Value to Argument Values

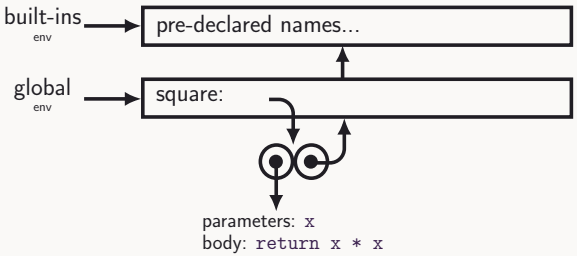
- Take the environment stored in the function object.
- Extend it with a call frame.
- Bind the parameters to the argument values.
- Bind each local variable, initially unassigned.
- Evaluate the function body in the resulting environment.

Example: square(5)



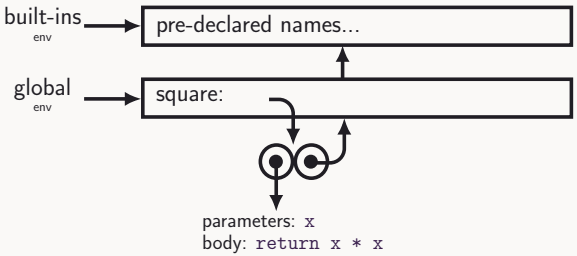
Example: square(5)

- Evaluate `square` and `5` in the program environment.



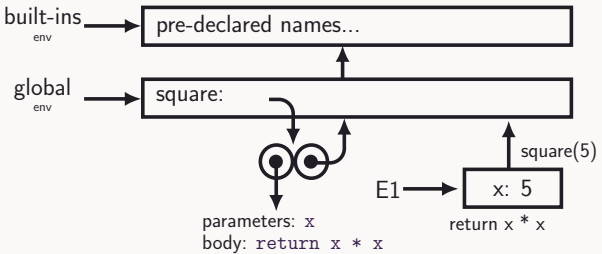
Example: square(5)

- Evaluate `square` and `5` in the program environment.
- Create a new frame that points to the environment of `square`.



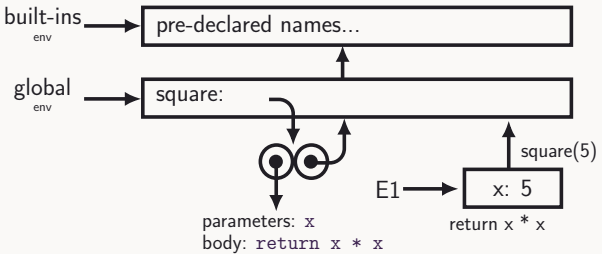
Example: square(5)

- Evaluate `square` and `5` in the program environment.
- Create a new frame that points to the environment of `square`.



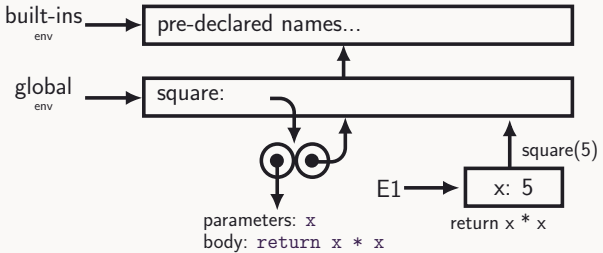
Example: square(5)

- Evaluate `square` and `5` in the program environment.
- Create a new frame that points to the environment of `square`.
- Bind the value 5 to parameter `x` in this frame.



Example: square(5)

- Evaluate `square` and `5` in the program environment.
- Create a new frame that points to the environment of `square`.
- Bind the value 5 to parameter `x` in this frame.
- Evaluate the body `return x * x`.



Example with a Local Variable

- The function `cube` has a local variable `y`.

```
def cube(x):
    y = x * x * x
    return y
print(cube(3))
```



cube Example

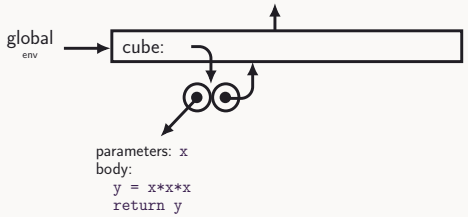
- After evaluating the definition of cube.

```
def cube(x):
    y = x * x * x
    return y
print(cube(3))
```



cube Example

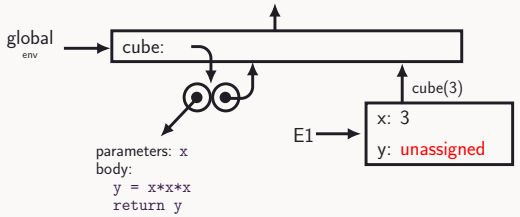
- After evaluating the definition of cube.




```
def cube(x):
    y = x * x * x
    return y
print(cube(3))
```

cube Example

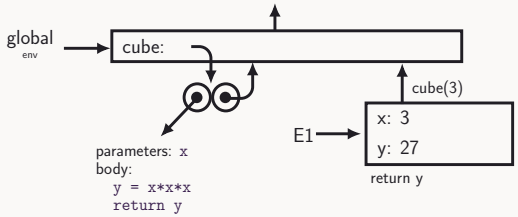
- Evaluate `cube(3)`, the argument of `print`.




```
def cube(x):
    y = x * x * x
    return y
print(cube(3))
```

cube Example

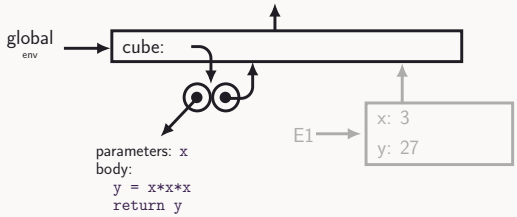
- Evaluate the function body.




```
def cube(x):
    y = x * x * x
    return y
print(cube(3))
```

cube Example

- Environment after evaluating `cube(3)`.




```
def cube(x):
    y = x * x * x
    return y
print(cube(3))
```

Example with Recursion

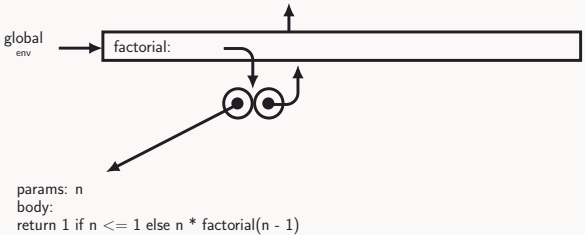
- A recursive definition of `factorial`:

```
def factorial(n):
    if n <= 1:
        return 1
    else:
        return n * factorial(n - 1)
print(factorial(2))
```



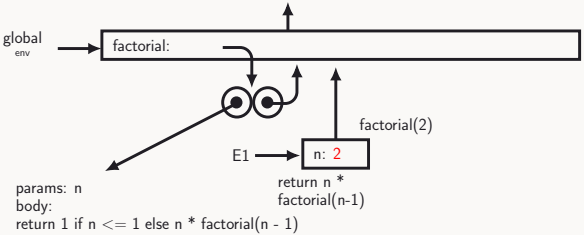
factorial Example

- After evaluating the definition of `factorial`.



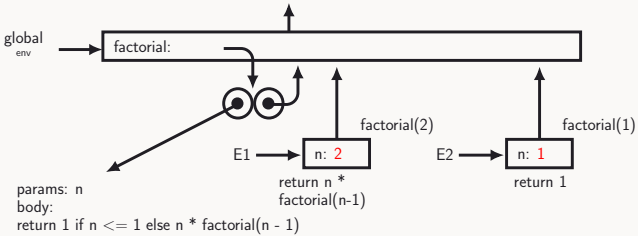
factorial Example

- Evaluate `factorial(2)`, the argument of `print`.



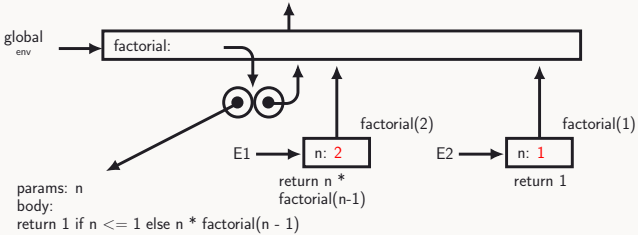
factorial Example

- Evaluate `factorial(n - 1)` in E1.



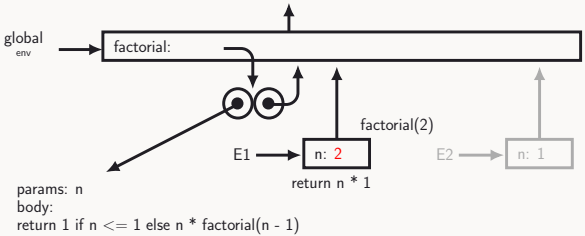
factorial Example

- Evaluate `return 1` in E2.



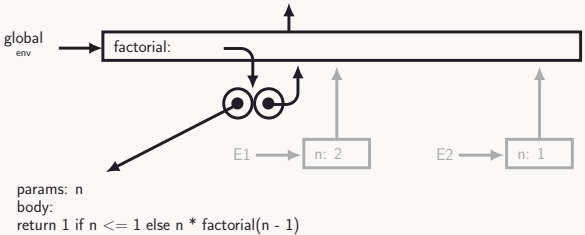
factorial Example

- Evaluate `return n * 1` in E1.



factorial Example

- Environment after evaluating `factorial(2)`.



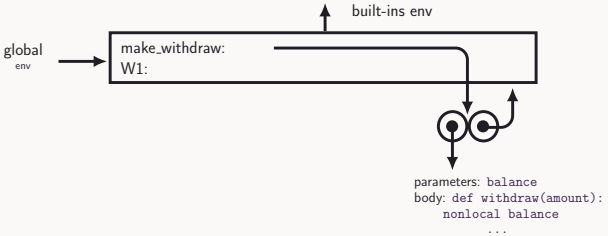
make_withdraw Example

```
def make_withdraw(balance):
    def withdraw(amount):
        nonlocal balance
        if balance >= amount:
            balance = balance - amount
            return balance
        return "Insufficient funds"
    return withdraw

W1 = make_withdraw(100)
print(W1(40))
W2 = make_withdraw(100)
```

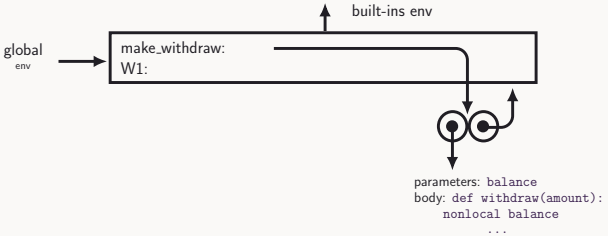
make_withdraw Example

- After defining `make_withdraw`.



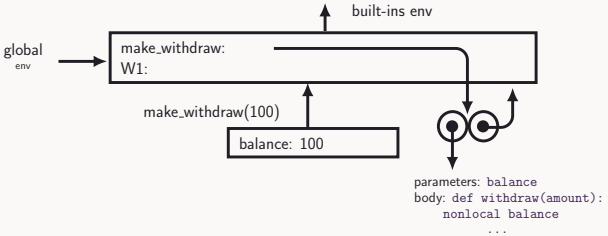
make_withdraw Example

- Evaluate `make_withdraw(100)`.



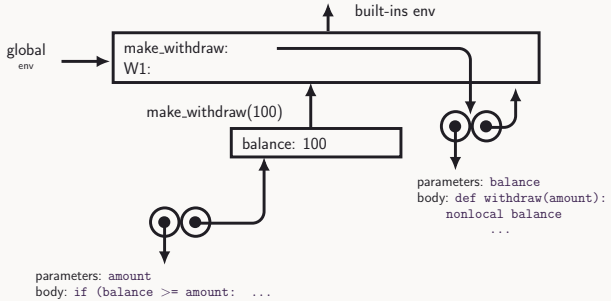
make_withdraw Example

- Evaluate `make_withdraw(100)`.



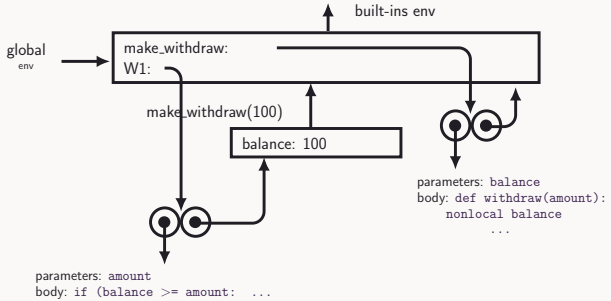
make_withdraw Example

- Evaluate `make_withdraw(100)`.



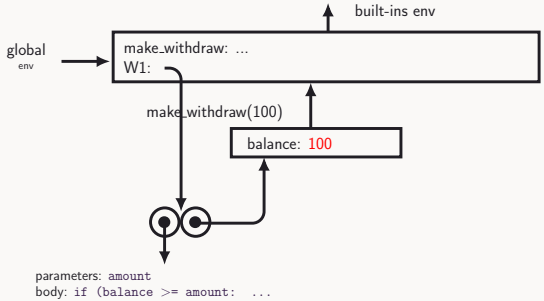
make_withdraw Example

- Evaluate `W1 = make_withdraw(100)`.



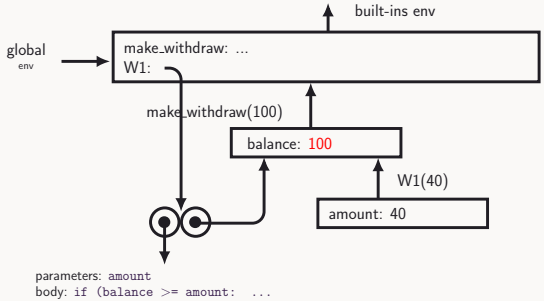
make_withdraw Example

- Evaluate `W1(40)`.



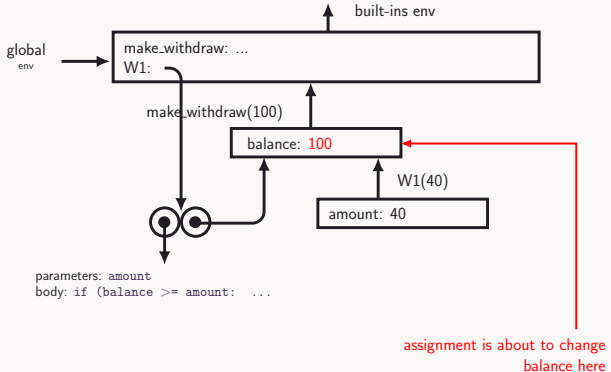
make_withdraw Example

- Evaluate `W1(40)`.



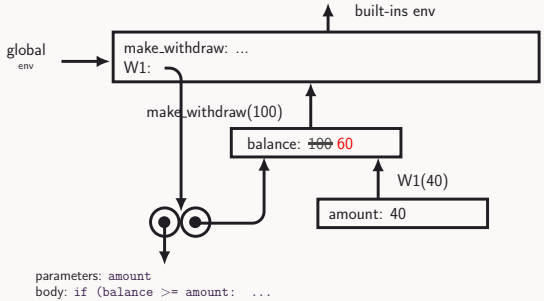
make_withdraw Example

- Evaluate `W1(40)`.



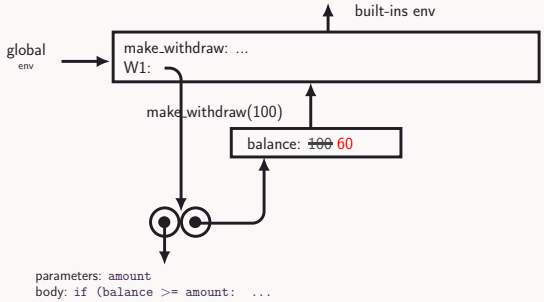
make_withdraw Example

- Evaluate `W1(40)`.



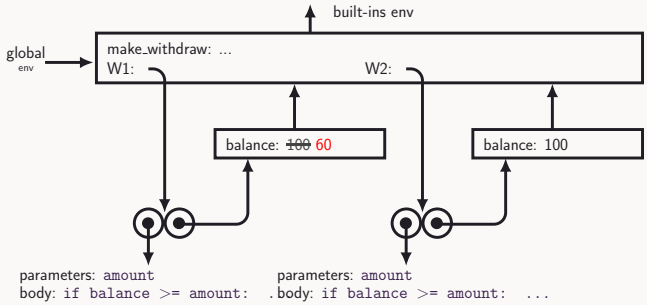
make_withdraw Example

- Environment after evaluating `W1(40)`.



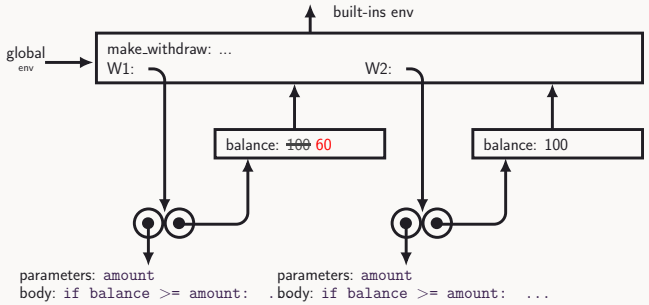
make_withdraw Example

- Environment after evaluating `W2 = make_withdraw(100)`.



make_withdraw Example

- Environment after evaluating `W2 = make_withdraw(100)`.
- `W2` has its own state variable.

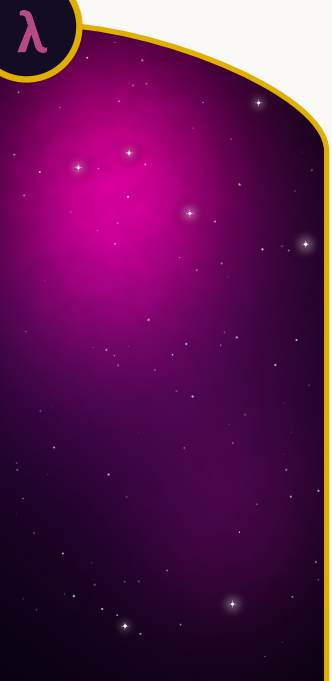


update Example

- What is the result of this program?



```
def update(x):
    x = x + 1
y = 2
update(y)
print(y)
```

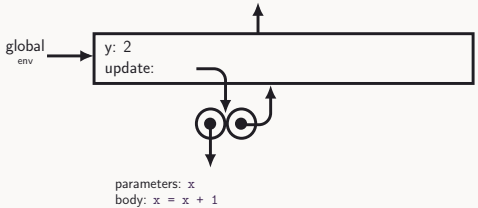


update Example

- After defining `update` and `y`.

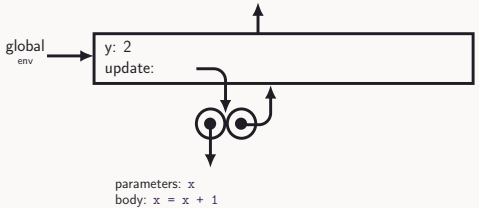
update Example

- After defining `update` and `y`.



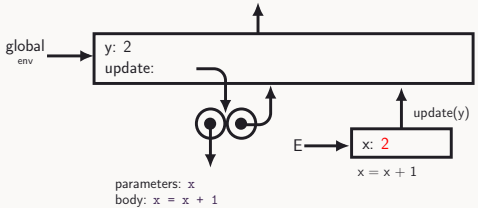
update Example

- Evaluate `update(y)` in the program environment.



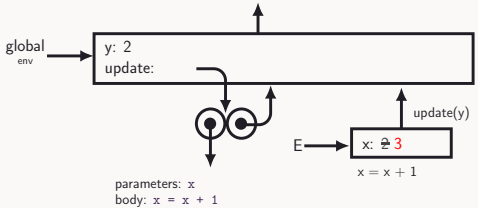
update Example

- Evaluate `update(y)` in the program environment.



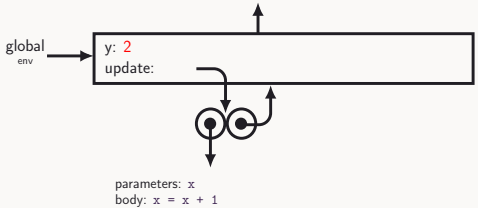
update Example

- Evaluate `x = x + 1` in E.



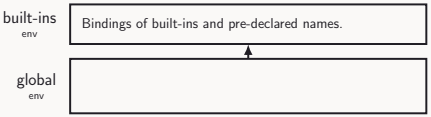
update Example

- Evaluate the argument `y` of `print(y)`.



The `global` Declaration

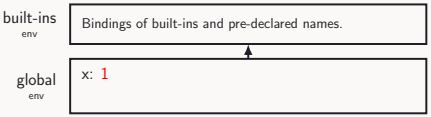
- What does this program print?




```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

The `global` Declaration

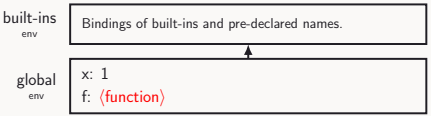
- `x = 1` binds `x` in the global frame.




```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

The `global` Declaration

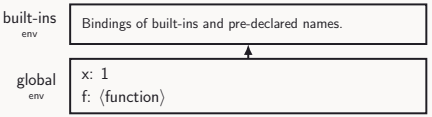
- Define `f`: the binding goes into the global frame.




```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

The `global` Declaration

- `print(x)` prints 1.

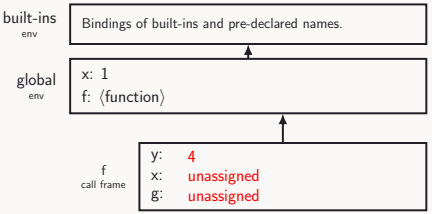



```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

output: 1

The `global` Declaration

- Evaluate `f(4)`: its frame has the parameter `y` and the local names `x` and `g`.

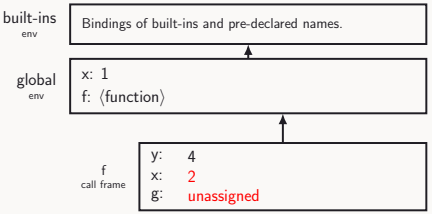



```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

output: 1

The `global` Declaration

- `x = 2` assigns the local `x` of `f`.

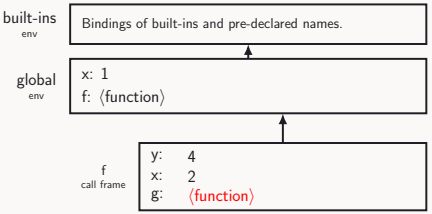



```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

output: 1

The `global` Declaration

- Define `g` in the frame of `f`.

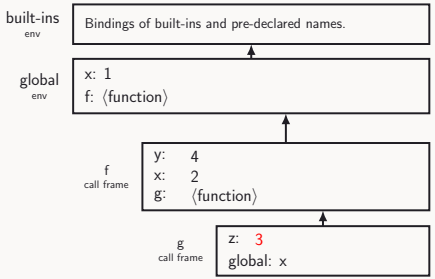



```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

output: 1

The `global` Declaration

- Evaluate `g(3)`: `x` is declared global in g, so it is not a local name of g.

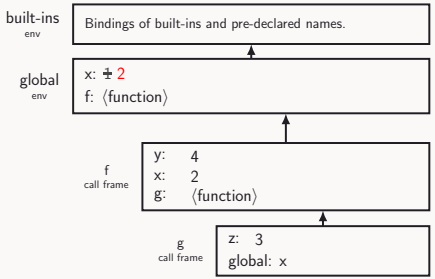



```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

output: 1

The `global` Declaration

- Because of `global x`, `x = x + 1` reads and assigns `x` in the global frame.

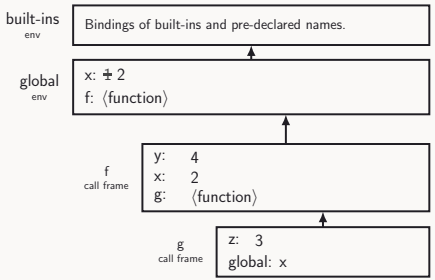



```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

output: 1

The `global` Declaration

- `print(x)` prints 2: the global `x` changed, the `x` of `f` did not.




```
x = 1
def f(y):
    x = 2
    def g(z):
        global x
        x = x + 1
    g(3)
print(x)
f(4)
print(x)
```

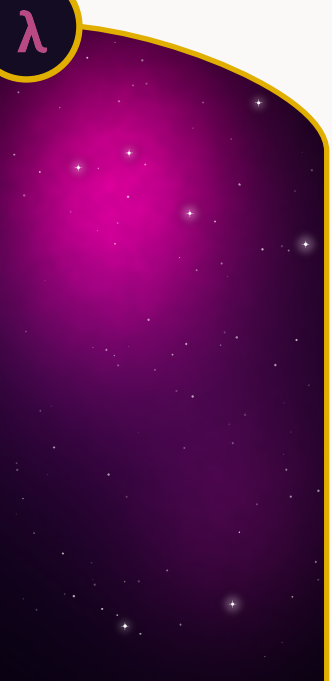
output: 1 2

pair_add_one Example

- What is the result of this program?

```
def pair_add_one(p):
    set_head(p, head(p) + 1)
    set_tail(p, tail(p) + 1)
q = pair(2, 5)
pair_add_one(q)
print(q)
```



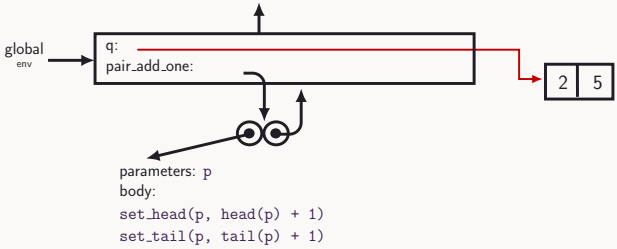


pair_add_one Example

- After defining `pair_add_one` and `q`.

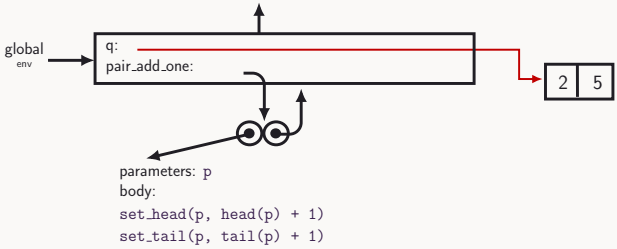
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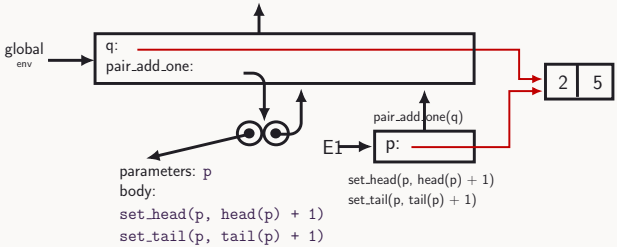
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- Evaluate `pair_add_one(q)`.



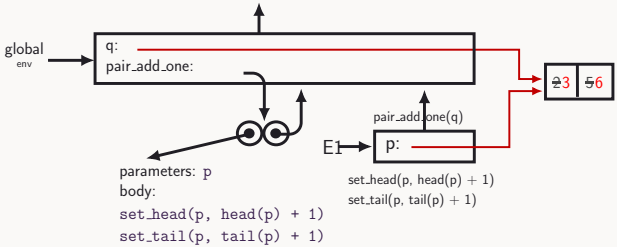
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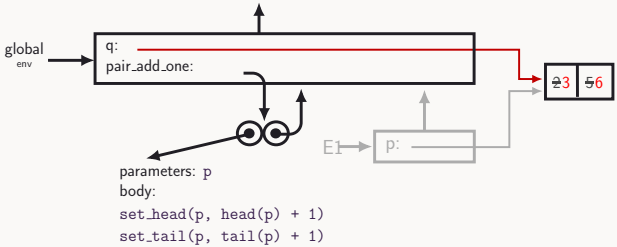
pair_add_one Example

- Evaluate the two mutator calls in E1.



pair_add_one Example

- Evaluate the argument `q` of `print(q)`.



Sharing and Identity

- `is` checks whether two names refer to the same pair.

```
a = pair(2, 3)
b = a
print(a is b)
```



```
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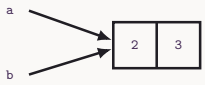


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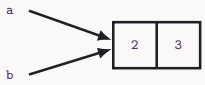
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```

```
True
```



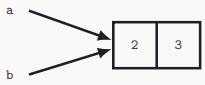
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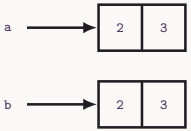
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```

True



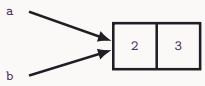


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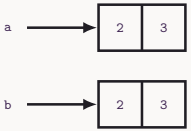
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b = a
print(a is b)
```

True




```
a = pair(2, 3)
b = pair(2, 3)
print(a is b)
```

False



Sharing and Identity

- Larger structures can share a pair.

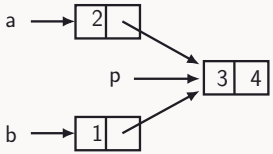
```
p = pair(3, 4)
a = pair(2, p)
b = pair(1, p)
print(tail(a) is tail(b))
```

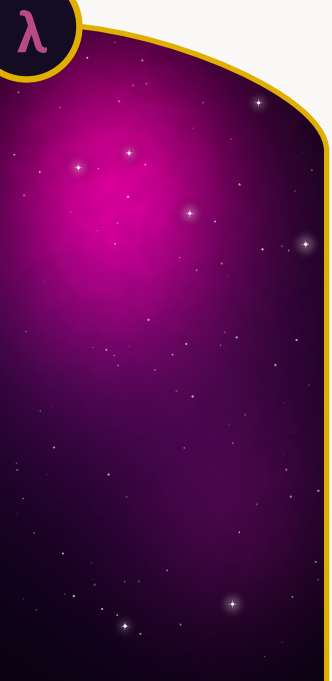


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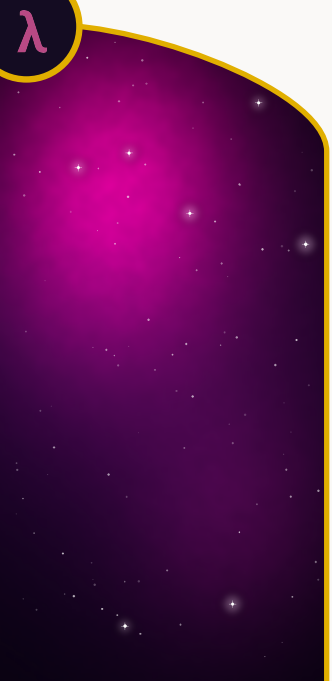
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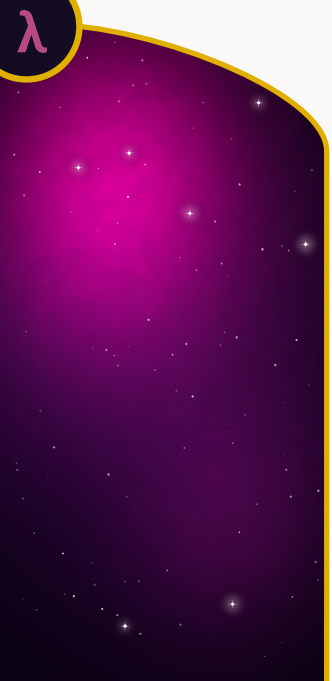


Identity in Python §3: is



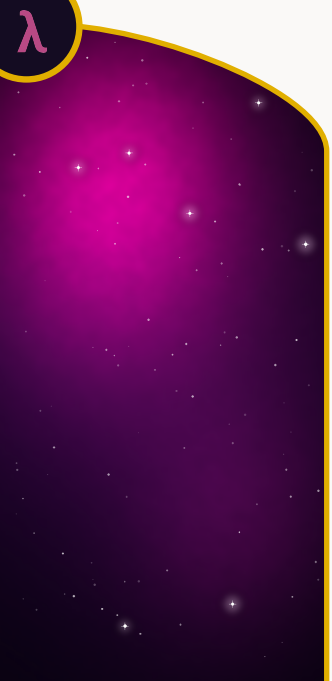
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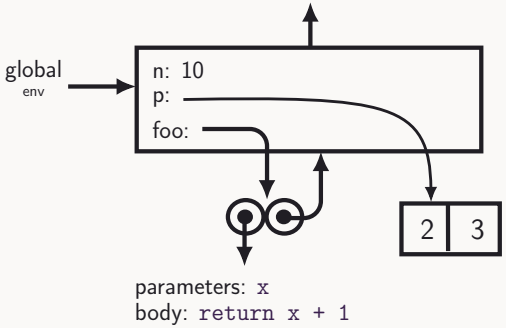
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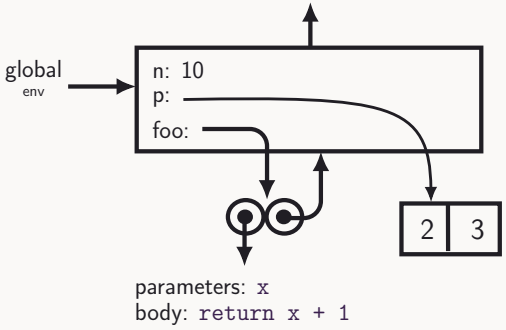
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  - Each list created by a list expression has a unique identity.

Drawing Compound Structures



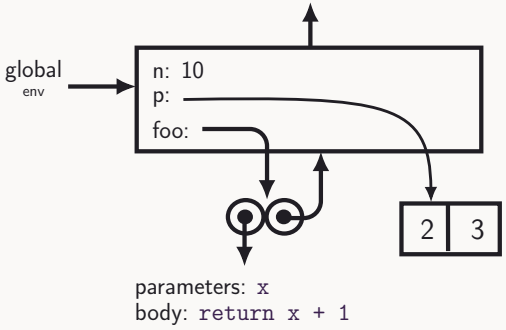
Drawing Compound Structures

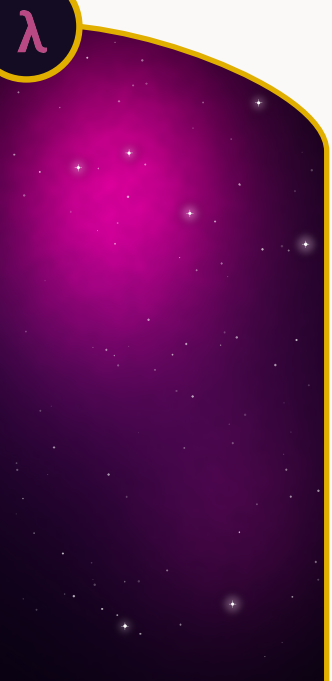
- Primitive values in bindings are drawn inside frames.



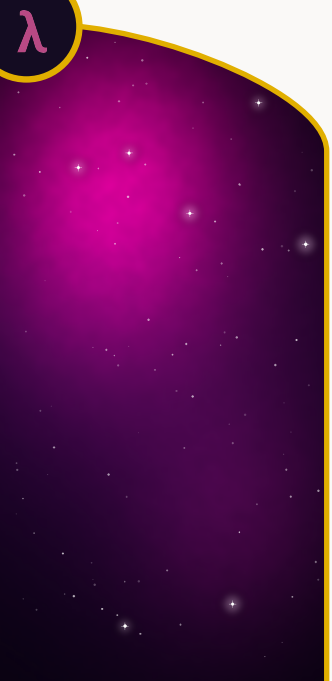
Drawing Compound Structures

- Primitive values in bindings are drawn inside frames.
- Bindings point to compound objects drawn outside the frames, making identity and sharing visible.



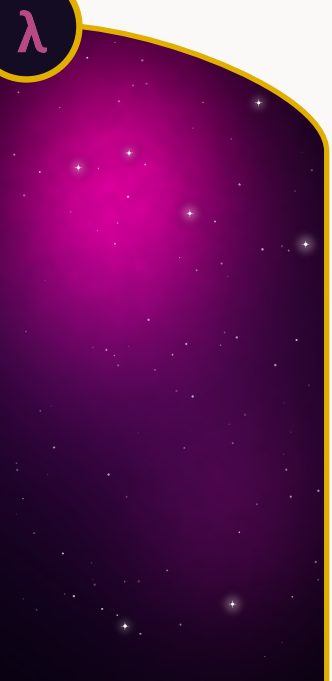


Summary of Environment Model



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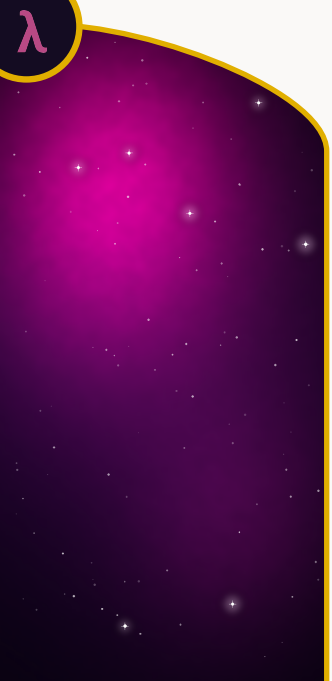
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- Bindings can point to compound objects; shared arrows represent shared identity.